

Exact Kalman Linear Attention: The Parallelization Obstruction

1 Exact Kalman update

Let $\tilde{\mathbf{S}}_t \in \mathbb{R}^{d \times d_v}$ denote the latent memory and $\mathbf{S}_t$ its filtered mean, with $\mathbf{k}_t, \mathbf{q}_t \in \mathbb{R}^d$ and $\mathbf{v}_t \in \mathbb{R}^{d_v}$. We consider the linear-Gaussian model

$$\tilde{\mathbf{S}}_t = \mathbf{D}_t \tilde{\mathbf{S}}_{t-1} + \mathbf{W}_t, \quad \mathbf{W}_t \sim \mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{\Omega}_t), \quad (1)$$

$$\mathbf{v}_t = \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, r_t \mathbf{I}), \quad r_t > 0. \quad (2)$$

We assume diagonal transition and process-noise covariance,

$$\mathbf{D}_t = \text{diag}(\mathbf{d}_t), \quad \mathbf{\Omega}_t = \text{diag}(\mathbf{\omega}_t), \quad \mathbf{\omega}_t \geq \mathbf{0}. \quad (3)$$

Here $\mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{\Omega}_t)$ means that the columns of $\mathbf{W}_t$ are independent $\mathcal{N}(\mathbf{0}, \mathbf{\Omega}_t)$ variables; process and observation noises are independent across time. We take $\mathbf{P}_0 \succ \mathbf{0}$ and assume each predictive covariance below is positive definite. The posterior covariance $\mathbf{P}_t \in \mathbb{R}^{d \times d}$ is nevertheless dense: conditioning on a general key couples its coordinates.

Given $(\mathbf{S}_{t-1}, \mathbf{P}_{t-1})$, the exact Kalman filter is

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \mathbf{\Omega}_t, \quad (4)$$

$$\boldsymbol{\kappa}_t = \frac{\hat{\mathbf{P}}_t \mathbf{k}_t}{r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t}, \quad \mathbf{S}_t = \hat{\mathbf{S}}_t + \boldsymbol{\kappa}_t (\mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{k}_t)^\top, \quad (5)$$

$$\mathbf{P}_t = \hat{\mathbf{P}}_t - \frac{(\hat{\mathbf{P}}_t \mathbf{k}_t)(\hat{\mathbf{P}}_t \mathbf{k}_t)^\top}{r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t}. \quad (6)$$

The corresponding linear-attention read is $\mathbf{o}_t = \mathbf{S}_t^\top \mathbf{q}_t$.

2 Information form

Define the posterior and predictive precision matrices

$$\mathbf{C}_t = \mathbf{P}_t^{-1}, \quad \hat{\mathbf{C}}_t = \hat{\mathbf{P}}_t^{-1}. \quad (7)$$

The Woodbury identity converts the rank-one covariance correction into the additive information update

$$\boxed{\mathbf{C}_t = \hat{\mathbf{C}}_t + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}} = \left(\mathbf{D}_t \mathbf{C}_{t-1}^{-1} \mathbf{D}_t^\top + \mathbf{\Omega}_t \right)^{-1} + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}. \quad (8)$$

Equivalently, with the whitened key $\bar{\mathbf{k}}_t = \mathbf{k}_t / \sqrt{r_t}$, the measurement contributes $\bar{\mathbf{k}}_t \bar{\mathbf{k}}_t^\top$.

Under the static, unit-noise specialization $\mathbf{D}_t = \mathbf{I}$, $\mathbf{\Omega}_t = \mathbf{0}$, and $r_t = 1$, this reduces to

$$\mathbf{C}_t = \mathbf{C}_{t-1} + \mathbf{k}_t \mathbf{k}_t^\top \quad (9)$$

If $\mathbf{\Omega}_t = \mathbf{0}$ and $\mathbf{D}_t$ is invertible, the more general update is

$$\mathbf{C}_t = \mathbf{D}_t^{-\top} \mathbf{C}_{t-1} \mathbf{D}_t^{-1} + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}. \quad (10)$$

With generic nonzero process noise, even diagonal $\mathbf{D}_t$ and $\mathbf{\Omega}_t$ leave the prediction in Equation (8) as a nonlinear dense matrix map.

3 Why parallelization remains difficult

The static case makes the distinction particularly clear. Define the information target $\mathbf{B}_t = \mathbf{C}_t \mathbf{S}_t$. Then

$$\mathbf{C}_t = \mathbf{C}_{t-1} + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}, \quad \mathbf{B}_t = \mathbf{B}_{t-1} + \frac{\mathbf{k}_t \mathbf{v}_t^\top}{r_t}. \quad (11)$$

Both are associative prefix sums. The difficult operation is the causal readout

$$\mathbf{S}_t = \mathbf{C}_t^{-1} \mathbf{B}_t, \quad \boldsymbol{\kappa}_t = \frac{\mathbf{C}_t^{-1} \mathbf{k}_t}{r_t}, \quad (12)$$

which requires a different dense solve at every prefix. Factoring only the final precision would expose earlier outputs to future keys. Sequential rank-one inverse updates have depth T . Factoring every prefix independently exposes concurrency, but repeats a dense factorization T times rather than sharing work through a useful prefix operator.

For the dynamic model, the problem is harder still: process noise makes the precision prediction itself non-additive. Thus the obstacle is not forming $\mathbf{k}_t \mathbf{k}_t^\top$, but applying a prefix-dependent dense inverse at every token while preserving causality. With invertible transitions the Riccati map admits a formal associative lifting to $2d \times 2d$ matrix-fractional maps, but each composition is a dense matrix operation and the products become numerically stiff under strong contraction. The algorithmic question is therefore not whether a formal parallel prefix exists, but whether its composition can be made inexpensive enough to replace the tokenwise chain.

4 Diagonal information and online variational inference

Suppose instead that the retained precision is diagonal,

$$\mathbf{C}_{t-1} = \text{diag}(\mathbf{c}_{t-1}), \quad \mathbf{C}_{t-1}^{-1} = \text{diag}(\mathbf{c}_{t-1}^{-1}). \quad (13)$$

Because $\mathbf{D}_t$ and $\mathbf{\Omega}_t$ are also diagonal, the predictive precision remains diagonal:

$$\hat{\mathbf{C}}_t = \text{diag}(\hat{\mathbf{c}}_t), \quad \hat{\mathbf{c}}_t = \frac{\mathbf{c}_{t-1}}{\mathbf{d}_t^2 + \boldsymbol{\omega}_t \odot \mathbf{c}_{t-1}}, \quad (14)$$

where all vector operations are elementwise. The exact gain under this predictive state is

$$\boldsymbol{\kappa}_t = \frac{\hat{\mathbf{c}}_t^{-1} \odot \mathbf{k}_t}{r_t + \sum_i k_{t,i}^2 / \hat{c}_{t,i}}, \quad (15)$$

so the uncertainty recurrence decomposes into independent scalar maps. All of their prefixes can be generated by an associative scan, after which the gain is evaluated tokenwise without a dense prefix solve.

The exact measurement update does not preserve this structure. By Equation (8), its precision is

$$\mathbf{C}_t^\star = \text{diag}(\widehat{\mathbf{c}}_t) + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}, \quad (16)$$

which is generally dense. We therefore project the exact one-step posterior π_t back to the diagonal family after every token:

$$q_t = \arg \min_{q=\mathcal{N}_{\text{col}}(\mathbf{M}, \text{diag}(\mathbf{c}^{-1}))} \text{KL}(q \parallel \pi_t). \quad (17)$$

The reverse-KL optimum preserves the exact posterior mean, $\mathbf{M} = \mathbf{S}_t$, and retains the diagonal of the exact posterior precision:

$$\boxed{\mathbf{c}_t = \widehat{\mathbf{c}}_t + \frac{\mathbf{k}_t \odot \mathbf{k}_t}{r_t}}. \quad (18)$$

Feeding $\text{diag}(\mathbf{c}_t^{-1})$ into the next prediction closes the online recursion. With $u_{t,i} = k_{t,i}^2/r_t$, each channel obeys

$$c_{t,i} = \frac{c_{t-1,i}}{d_{t,i}^2 + \omega_{t,i} c_{t-1,i}} + u_{t,i}. \quad (19)$$

This is a scalar Möbius map, whose token-local 2×2 representations compose associatively. Online variational inference therefore replaces each prefix-dependent dense inverse with d independent scalar scans while retaining the exact Kalman mean update under the diagonal predictive prior.

5 Oracle-free gain approximation

The gain need not be computed by first forming a covariance inverse. Combining the information update with the Kalman gain gives the equivalent linear system

$$\boxed{\mathbf{C}_t \boldsymbol{\kappa}_t = \frac{\mathbf{k}_t}{r_t}, \quad \mathbf{C}_t = \widehat{\mathbf{C}}_t + \frac{\mathbf{k}_t \mathbf{k}_t^\top}{r_t}}. \quad (20)$$

This suggests approximating the solution of a known positive-definite system, rather than defining an approximation in terms of the unknown exact gain.

5.1 Diagonal-Kalman-compatible elementwise gains

Consider

$$\boldsymbol{\kappa}_t(\mathbf{b}) = \mathbf{b} \odot \mathbf{k}_t. \quad (21)$$

Let $\mathcal{I}_t = \{i : k_{t,i} \neq 0\}$ and set $b_i = 0$ outside $\mathcal{I}_t$. For a positive diagonal predictive precision, Equation (15) gives

$$b_i = \frac{\widehat{c}_{t,i}^{-1}}{r_t + \zeta_t}, \quad \zeta_t = \sum_{j \in \mathcal{I}_t} \frac{k_{t,j}^2}{\widehat{c}_{t,j}}. \quad (22)$$

Thus $b_i \geq 0$, and the effective write along the current key is

$$\beta_t^{\text{eff}} := \mathbf{k}_t^\top \boldsymbol{\kappa}_t = \sum_{i \in \mathcal{I}_t} b_i k_{t,i}^2 = \frac{\zeta_t}{r_t + \zeta_t} \in [0, 1). \quad (23)$$

If $\hat{\mathbf{y}}_t = \hat{\mathbf{S}}_t^\top \mathbf{k}_t$ is the value recalled before the update, then

$$\mathbf{S}_t^\top \mathbf{k}_t = \hat{\mathbf{y}}_t + \beta_t^{\text{eff}}(\mathbf{v}_t - \hat{\mathbf{y}}_t), \quad (24)$$

$$\mathbf{v}_t - \mathbf{S}_t^\top \mathbf{k}_t = (1 - \beta_t^{\text{eff}})(\mathbf{v}_t - \hat{\mathbf{y}}_t). \quad (25)$$

The first constraint below prevents a channel from writing opposite to its key coordinate, while the second bounds the total same-key write:

$$\mathcal{B}_t := \left\{ \mathbf{b} : b_i \geq 0 \ (i \in \mathcal{I}_t), \quad \sum_{i \in \mathcal{I}_t} b_i k_{t,i}^2 \leq 1 \right\}. \quad (26)$$

The finite positive-precision family satisfies strict inequalities. The weak inequalities include its closure, and every interior point corresponds to the positive diagonal predictive precision

$$\hat{c}_i = \frac{1 - \sum_{j \in \mathcal{I}_t} b_j k_{t,j}^2}{r_t b_i}, \quad i \in \mathcal{I}_t. \quad (27)$$

There is an important limitation: if every key coordinate is nonzero, the map $\mathbf{b} \mapsto \mathbf{b} \odot \mathbf{k}_t$ is a bijection. An unrestricted channelwise $\mathbf{b}$ can therefore represent any gain and is no easier to find than the original dense solve. Computational savings require either a lower-dimensional structure on $\mathbf{b}$ or a fixed-work approximate solver.

5.2 A computable variational problem

Let $\mathbf{K}_t = \text{diag}(\mathbf{k}_t)$ and $\mathbf{s}_t = \mathbf{k}_t \odot \mathbf{k}_t$. Restricting the quadratic variational form of Equation (20) to $\kappa_t = \mathbf{K}_t \mathbf{b}$ gives

$$\boxed{\mathbf{b}_t^\dagger \in \arg \min_{\mathbf{b} \in \mathcal{B}_t} \Phi_t(\mathbf{b}), \quad \Phi_t(\mathbf{b}) = \frac{r_t}{2} \mathbf{b}^\top \mathbf{K}_t \mathbf{C}_t \mathbf{K}_t \mathbf{b} - \mathbf{s}_t^\top \mathbf{b}.} \quad (28)$$

Unlike an oracle projection, this objective contains only the available precision, key, and observation noise. Its gradient is

$$\nabla \Phi_t(\mathbf{b}) = r_t \mathbf{k}_t \odot [\mathbf{C}_t(\mathbf{b} \odot \mathbf{k}_t)] - \mathbf{s}_t. \quad (29)$$

Without the constraints and with a full-support key, its stationarity equation is exactly Equation (20); with the constraints it is the $\mathbf{C}_t$ -energy projection of that system onto the capped diagonal-compatible gain set. This is a computable energy surrogate, not the constrained posterior-MSE projection: the two objectives share the exact unconstrained minimizer but use different projection metrics. Evaluating Equation (28) never requires $\hat{\mathbf{C}}_t^{-1} \mathbf{k}_t$.

A low-dimensional coefficient family. One explicit structural restriction is

$$\mathbf{b} = \mathbf{R}_t \boldsymbol{\theta}, \quad \mathbf{R}_t \in \mathbb{R}^{d \times m}, \quad m \ll d, \quad (30)$$

where columns of $\mathbf{R}_t$ can, for example, tie channels into fixed groups. The computable problem becomes the m -dimensional quadratic program

$$\boldsymbol{\theta}_t^\dagger \in \arg \min_{\substack{\mathbf{R}_t \boldsymbol{\theta} \geq \mathbf{0} \\ \mathbf{s}_t^\top \mathbf{R}_t \boldsymbol{\theta} \leq 1}} \left\{ \frac{1}{2} \boldsymbol{\theta}^\top \mathbf{H}_t \boldsymbol{\theta} - \mathbf{h}_t^\top \boldsymbol{\theta} \right\}, \quad (31)$$

where

$$\mathbf{H}_t = r_t \mathbf{R}_t^\top \mathbf{K}_t \mathbf{C}_t \mathbf{K}_t \mathbf{R}_t, \quad \mathbf{h}_t = \mathbf{R}_t^\top \mathbf{s}_t. \quad (32)$$

This needs m precision–vector products followed by a small local solve. For $\mathbf{k}_t \neq \mathbf{0}$, taking $m = 1$ and $\mathbf{R}_t = \mathbf{1}$ gives $\boldsymbol{\kappa}_t = \beta_t \mathbf{k}_t$ with the closed form

$$\boxed{\beta_t = \frac{\|\mathbf{k}_t\|_2^2}{r_t \mathbf{k}_t^\top \mathbf{C}_t \mathbf{k}_t}, \quad \mathbf{k}_t^\top \boldsymbol{\kappa}_t = \frac{\|\mathbf{k}_t\|_2^4}{r_t \mathbf{k}_t^\top \hat{\mathbf{C}}_t \mathbf{k}_t + \|\mathbf{k}_t\|_2^4} < 1.} \quad (33)$$

The first expression uses the posterior precision $\mathbf{C}_t$; the second uses $\mathbf{C}_t = \hat{\mathbf{C}}_t + \mathbf{k}_t \mathbf{k}_t^\top / r_t$ to expose the same-key bound. A nested sequence of column spaces for $\mathbf{R}_t$ interpolates between this scalar gain and the full channelwise problem. Setting $m = d$ and $\mathbf{R}_t = \mathbf{I}$ removes the structural saving and recovers the full constrained channelwise problem; it equals the dense Kalman gain only when that gain belongs to $\mathcal{B}_t$.

A fixed-work channelwise solver. Alternatively, retain one coefficient per channel but run exactly L projected-gradient steps,

$$\mathbf{b}_t^{(\ell+1)} = \Pi_{\mathcal{B}_t} \left(\mathbf{b}_t^{(\ell)} - \eta_{t,\ell} \left\{ r_t \mathbf{k}_t \odot [\mathbf{C}_t (\mathbf{b}_t^{(\ell)} \odot \mathbf{k}_t)] - \mathbf{s}_t \right\} \right), \quad \ell = 0, \dots, L-1. \quad (34)$$

The projection onto $\mathcal{B}_t$ is a nonnegative weighted-simplex projection, implemented by a scalar bisection or sorting. A sufficient fixed step-size condition is

$$0 < \eta_{t,\ell} \leq \lambda_{\max}^{-1}(r_t \mathbf{K}_t \mathbf{C}_t \mathbf{K}_t), \quad (35)$$

with the matrices restricted to the support of $\mathbf{k}_t$; a monotone line search is another option. Equation (22) provides a natural diagonal-variational initialization for $\mathbf{b}_t^{(0)}$. The integer L is now an explicit speed–accuracy knob: every iteration uses one dense matrix–vector product, reductions, and elementwise operations, but no inverse and no oracle gain.

5.3 Parallel scope

In the static model, all posterior precisions are available from the associative prefix

$$\mathbf{C}_t = \mathbf{C}_0 + \sum_{\tau \leq t} \frac{\mathbf{k}_\tau \mathbf{k}_\tau^\top}{r_\tau}. \quad (36)$$

At projected-gradient iteration ℓ , the required product for every token can equivalently be written

$$\mathbf{C}_t \mathbf{u}_t^{(\ell)} = \mathbf{C}_0 \mathbf{u}_t^{(\ell)} + \sum_{\tau \leq t} \frac{\mathbf{k}_\tau (\mathbf{k}_\tau^\top \mathbf{u}_t^{(\ell)})}{r_\tau}, \quad \mathbf{u}_t^{(\ell)} = \mathbf{b}_t^{(\ell)} \odot \mathbf{k}_t. \quad (37)$$

Thus all positions can execute each solver stage concurrently. Materializing all dense prefix matrices once gives sequence depth $O(\log T + L)$, after which the L matrix–vector steps are token-local. Alternatively, Equation (37) is a causal linear-attention contraction with query $\mathbf{u}_t^{(\ell)}$; recomputing it at every stage gives $O(L \log T)$ sequence depth without materializing every prefix matrix. In either schedule, the construction replaces each dense inverse by a fixed number of parallel precision–vector products. It does not remove dense arithmetic, and with nonzero process noise constructing the exact dense $\mathbf{C}_t$ prefixes still has the nonlinear Riccati obstruction in Equation (8).